Machine Learning for Urban Air Pollutant (PM 2.5

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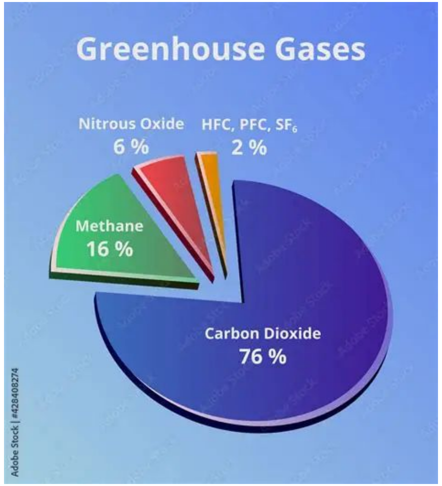
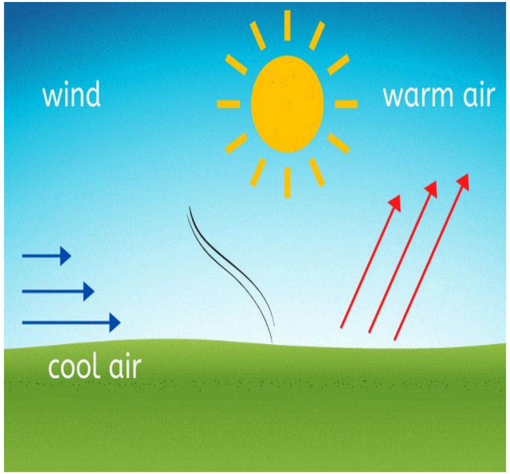
PM2.5 Mathematical model

Physics-Inspired PM2.5 Model

We combine emissions, atmospheric chemistry, and meteorology into a generalized formulation:

$$\mathrm{PM}_{2.5} = \underbrace{\alpha_1 E}_{\text{primary emissions}} + \underbrace{\alpha_2(SO_2 + NO_x + VOCs)}_{\text{secondary formation}} + \underbrace{\alpha_3 \cdot RH}_{\text{humidity effect}} + \underbrace{\alpha_4 \cdot T}_{\text{temperature effect}} - \underbrace{\alpha_5 \cdot W}_{\text{wind dispersion}} + \epsilon$$





Baseline rolling mean of PM2.5

3-Day Rolling Mean Baseline Equation

$$\mathrm{PM2.5}_t \approx \frac{\mathrm{PM2.5}_{t-1} + \mathrm{PM2.5}_{t-2} + \mathrm{PM2.5}_{t-3}}{3}$$

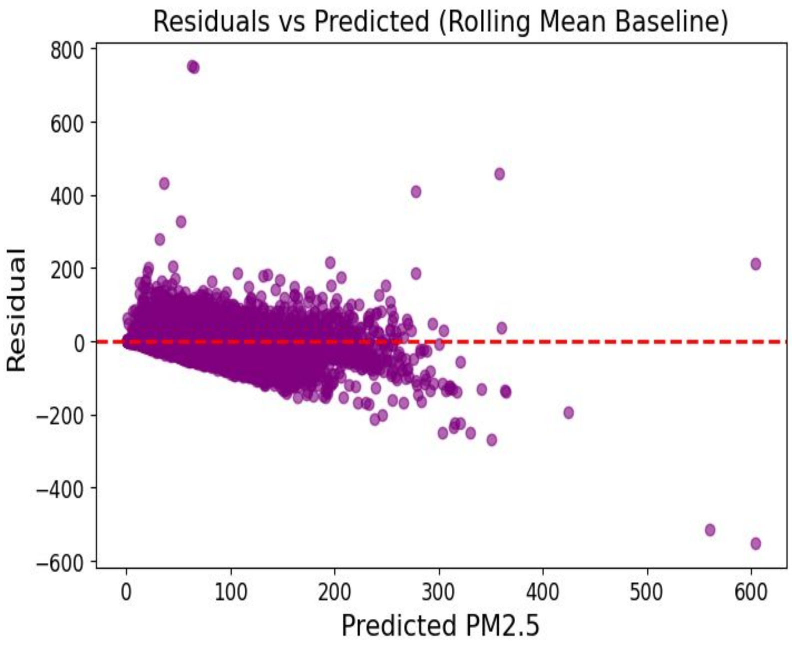
Where:

- $\mathrm{PM2.5}_t$ → Predicted PM2.5 concentration on day *t*
- $\mathrm{PM2.5}_{t-1}, \mathrm{PM2.5}_{t-2}, \mathrm{PM2.5}_{t-3}$ → Observed PM2.5 concentrations from the previous 3 days

Note: This formula **smooths short-term fluctuations** and captures temporal trends but does not include meteorology or emissions.

Residuals vs Predicted PM2.5

- **When the baseline predicts high PM2.5, the actual values tend to be lower.** This is a sign of *systematic bias*.
- **High variance at low predicted values**
- **Heteroscedasticity** The spread of residuals changes with predicted value.



Rolling Mean (3-day lag) Baseline Metrics: **RMSE: 28.0360**

A simple rolling mean cannot capture the dynamics of data.

1. Random Forest Interpretation

Model Form (Conceptual)

$$\mathrm{PM}_{2.5} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{Tree}_i(X)$$

- Each tree learns **rules** like:
  - *If humidity is high and wind is low → PM2.5 increases*
  - *If NO_2 is high → pollution likely high*

2. XGBoost Interpretation

Model Form (Sequential Learning)

$$\mathrm{PM}_{2.5} = \sum_{m=1}^{M} f_m(X)$$

- Each new tree **corrects previous errors**

3. Gradient Boosting Interpretation

Model Form

$$\mathrm{PM}_{2.5} = \sum_{m=1}^{M} \gamma_m \cdot h_m(X)$$

- Trees are added **gradually to reduce error**

Linear Regression

Physics-Inspired PM2.5 Model (Refined)

1. Core Equation

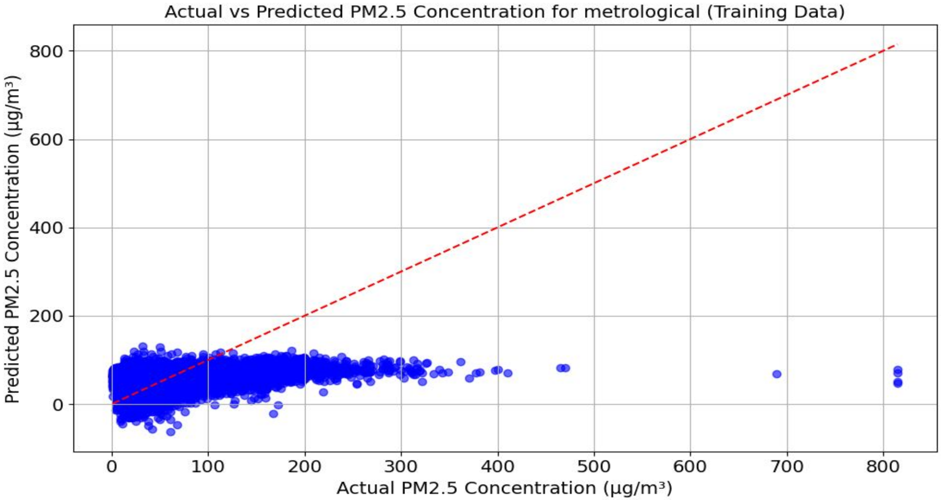
$$\mathrm{PM}_{2.5} = \alpha_1 E + \alpha_2(SO_2 + NO_x + VOCs) + \alpha_3 RH + \alpha_4 T - \alpha_5 W + \epsilon$$

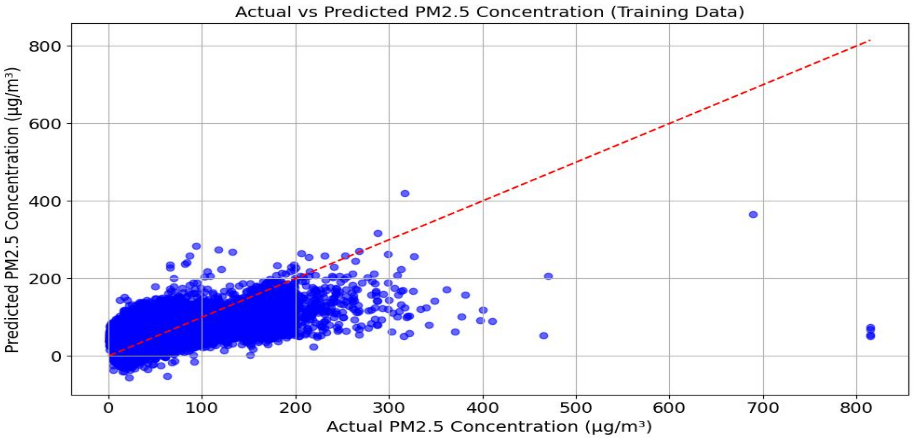
Key improvement:

In practice, this can also be written in a more **statistically usable linear form:**

$$\mathrm{PM}_{2.5} = \beta_0 + \beta_1 E + \beta_2 SO_2 + \beta_3 NO_x + \beta_4 VOCs + \beta_5 RH + \beta_6 T - \beta_7 W + \epsilon$$

This version maps **directly to Linear Regression / ML models.**



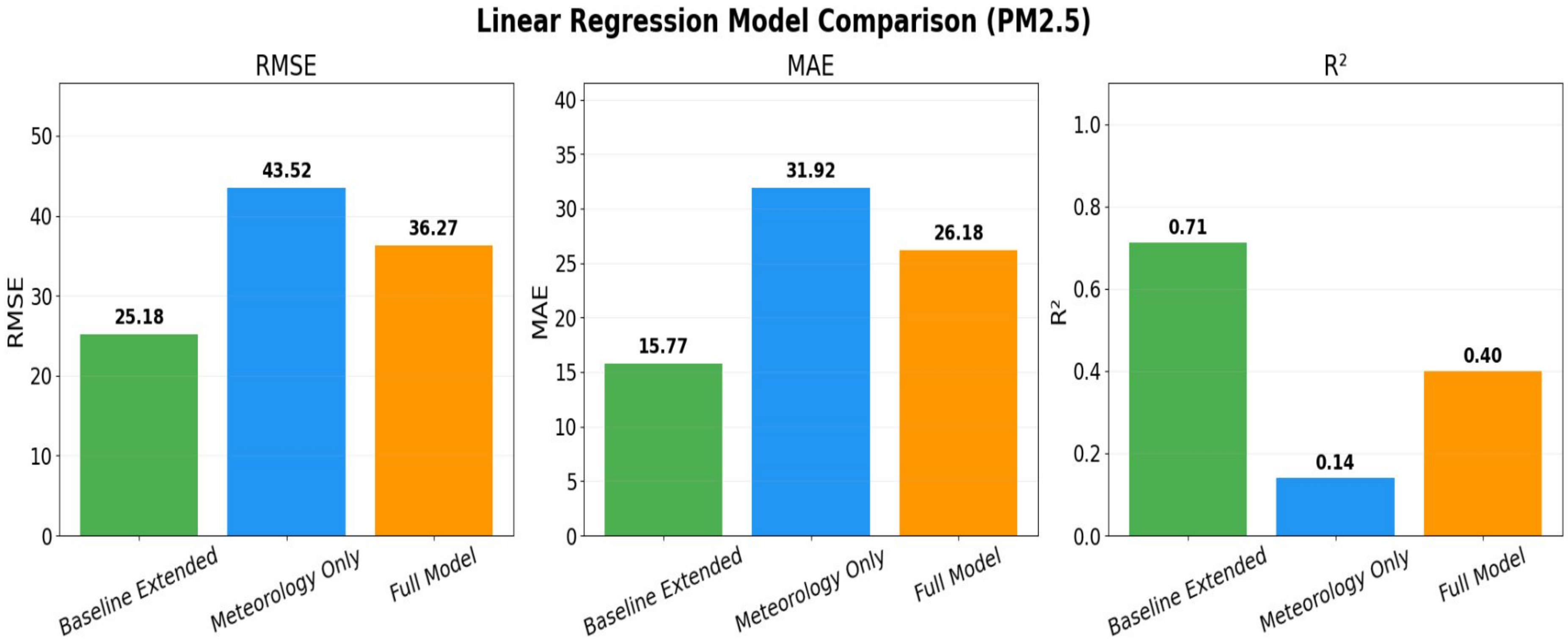


This is exactly what happens when:

- The true function is nonlinear
- But the model is forced to fit a straight line

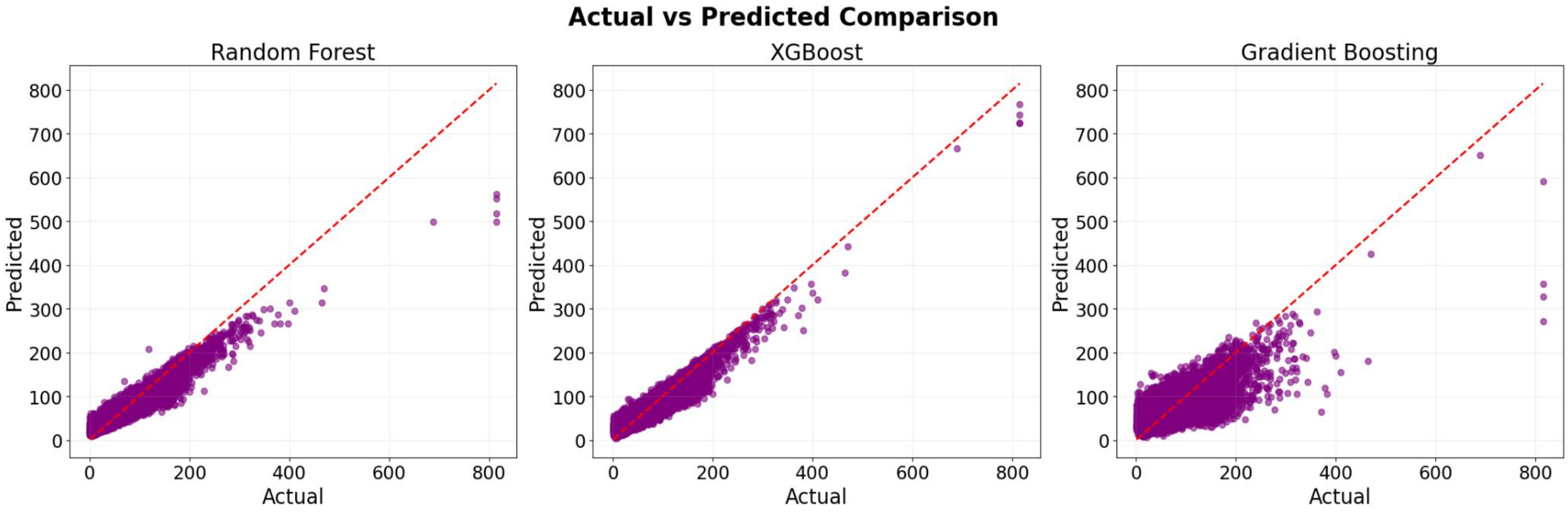
Linear regression tries its best, but PM2.5 dynamics are rarely linear.

Linear Regression



The extended baseline model performs better compare to Baseline model with RMSE = 28.03

RandomForestRegression, XGBoost and Gradient Boosting



Random Forest is **very good at capturing nonlinear relationships**, but it can be slightly noisy because each tree is independent. It tends to:

- Capture local patterns well
- Handle meteorological interactions

XGBoost is usually **the strongest model** for this kind of environmental regression task because:

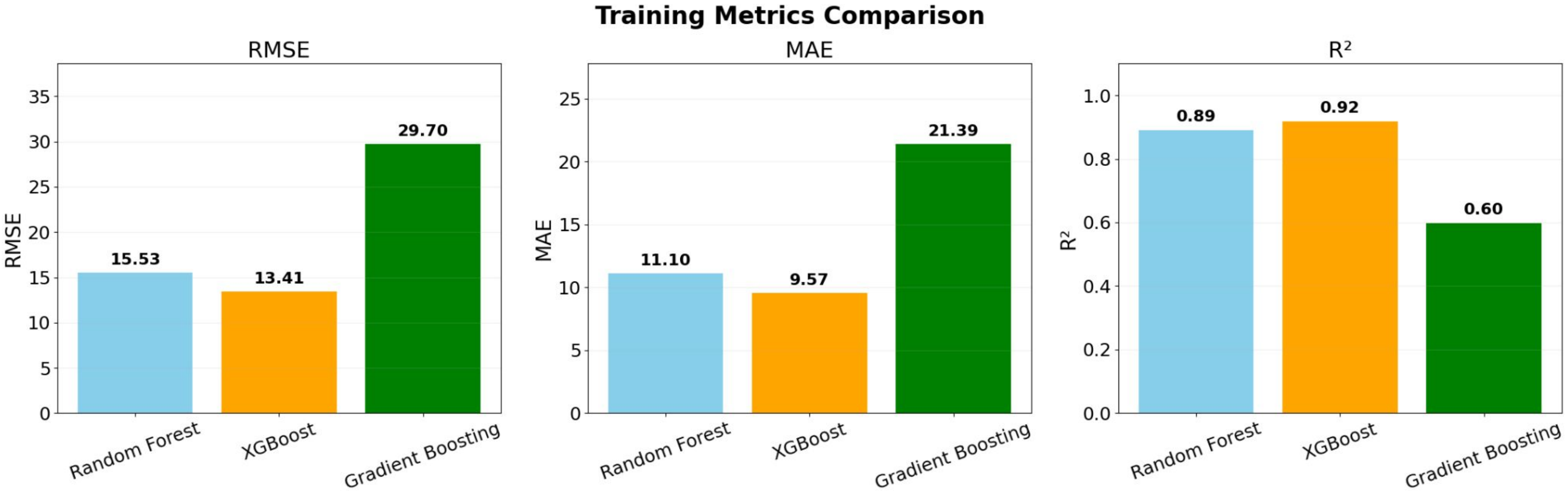
- It handles nonlinearities
- It reduces bias and variance through boosting
- It adapts to rare events better than Random Forest

The sklearn GradientBoostingRegressor is solid but:

- Less optimized
- Less flexible
- More prone to underfitting compared to XGBoost

It’s good, but not as sharp as XGBoost.

Evaluation Metrics

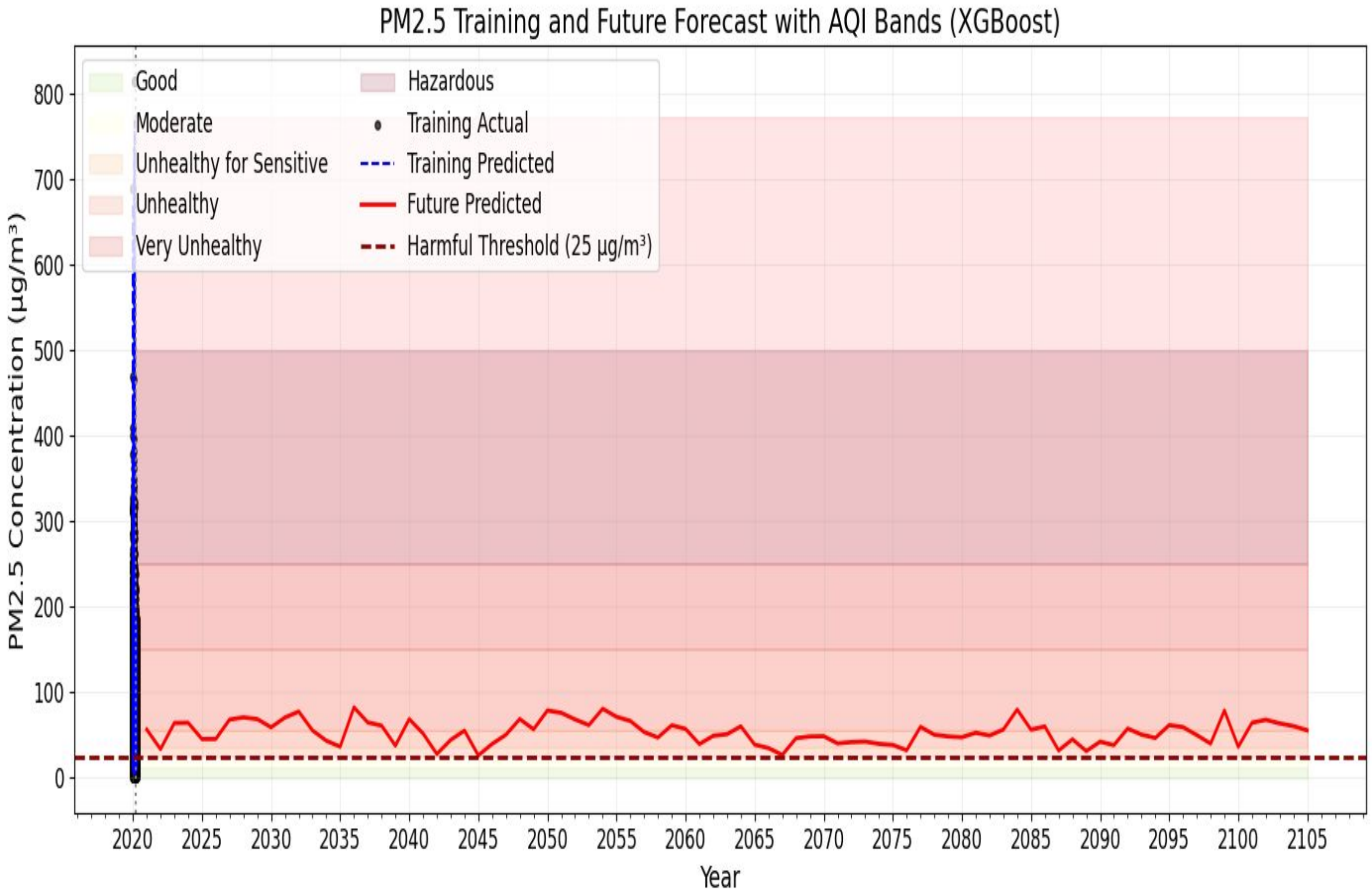


XGBoost is best among all the models with RMSE = 13.41

Translating Predictions to Community Impact

Air Quality Levels (Business performance) Translate PM2.5 µg/m³ values into categories:

- **Good (0–12)**
- **Moderate (12–35)**
- **Unhealthy for sensitive groups (35–55)**
- **Unhealthy (55–150)**
- **Very unhealthy (>150)**



Conclusion

- Tree-based ensemble models such as Random Forest, Gradient Boosting, and XGBoost extend the linear physics-inspired formulation of PM2.5 by capturing nonlinear interactions, threshold effects, and complex atmospheric processes that are not represented in traditional linear regression.

- The Baseline model fail to capture dynamics relate to PM2.5.

- The runtime of model increase with increase in number of feature.

- PM2.5 is at alarming level for human health, its effect is already seen in Global south. Strictly reducing emission is the only solution.

Future Work

- Parallelization version would be implemented in order for model to run fast.

- Hybrid model can be use in order to get better performance.

Value of the Product : Predict air quality levels and empower communities to plan and protect their health

Empowering Communities

- **Participatory monitoring:** Encourage citizens to use low-cost PM2.5 sensors.
- **Policy influence:** Provide aggregated data to local authorities for traffic/industrial regulation.
- **Health planning:** Track long-term exposure for healthcare decisions.
- **Community awareness campaigns:** Use predicted air quality data to inform schools, clinics, and neighborhoods.

References:

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2. A. Engels, J. Marotzke, E. G. Gresse, A. López-Rivera, A. Pagnone, and J. Wilkens (eds.), *Hamburg Climate Futures Outlook 2023: The Plausibility of a 1.5°C Limit to Global Warming—Social Drivers and Physical Processes*. Hamburg, Germany: Cluster of Excellence Climate, Climate Change, and Society (CLICCS), 2023.


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